<u>Last time</u> $\lim_{n\to\infty} a_n = a \iff \forall \varepsilon>0 \ \exists N\in\mathbb{N}$ s.t. $|a_n - a| < \varepsilon \quad \forall n \geq N$

E.g. $\lim_{n\to\infty} \frac{1}{n} = 0$, just take $N > \frac{1}{\varepsilon}$.

<u>Q</u>: How can we prove $\lim_{n\to\infty} \frac{n^3+5n+8}{3n^3+2n^2+7} = \frac{1}{3}$?

<u>Limit laws</u>: If $\lim_{n\to\infty} a_n$ and $\lim_{n\to\infty} b_n$ exist, then:

① $\lim_{n\to\infty} ka = k \cdot \lim_{n\to\infty} a_n \quad \forall k\in\mathbb{R}$

② $\lim_{n\to\infty} (a_n + b_n) = \lim_{n\to\infty} a_n + \lim_{n\to\infty} b_n$

Last time

③ $\lim_{n\to\infty} (a_n b_n) = (\lim_{n\to\infty} a_n)\cdot(\lim_{n\to\infty} b_n)$ ← Next

④ $\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n}$, if also $\lim_{n\to\infty} b_n \neq 0$ and $b_n \neq 0 \ \forall n \geq 1$.

<u>Thm</u> If the sequences $\{a_n\}_{n\geq 1}$ and $\{b_n\}_{n\geq 1}$ converge to a and b respectively, then $\{a_n b_n\}_{n\geq 1}$ converges and

$$\lim_{n\to\infty} (a_n b_n) = ab.$$

<u>Pf</u>: Fix $\varepsilon > 0$. As $\lim_{n\to\infty} a_n = a$, $\exists N_1 \in \mathbb{N}$ s.t.

$$|a_n - a| < \frac{\varepsilon}{2(|b|+1)} \quad \forall n \geq N_1.$$

As $\lim_{n\to\infty} b_n = b$, $\exists N_2 \in \mathbb{N}$ s.t.

$$|b_n - b| < \min\left\{\frac{\varepsilon}{2(|a|+1)}, 1\right\} \quad \forall n \geq N_2.$$

Set $N = \max\{N_1, N_2\}$. Then, for $n \geq N$ we have

$$
\begin{aligned}
|a_n b_n - ab| &= |a_n b_n - a b_n + a b_n - ab| \\
&\leq |a_n b_n - a b_n| + |a b_n - ab| \\
&= |a_n - a| \cdot |b_n| + |a| \cdot |b_n - b| \\
&\leq \frac{\varepsilon}{2} \cdot \underbrace{\frac{|b_n|}{|b|+1}} + \frac{|a|}{|a|+1} \cdot \frac{\varepsilon}{2} \\
&< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.
\end{aligned}
$$

$$|b_n - b| < 1 \implies |b_n| \leq |b_n - b| + |b| < 1 + |b|$$

$$\implies \frac{|b_n|}{|b|+1} < 1$$

So $\lim_{n\to\infty} (a_n b_n) = ab$. $\square$

<u>Thm</u> If the sequences $\{a_n\}_{n\geq 1}$ and $\{b_n\}_{n \geq 1}$ converge to a and b respectively, $b \neq 0$, and $b_n \neq 0$ $\forall n \geq 1$, then $\{\frac{a_n}{b_n}\}_{n\geq 1}$ converges and

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{a}{b}.$$

<u>Pf</u>: It suffices to prove $\lim_{n\to\infty} \frac{1}{b_n} = \frac{1}{b}$. Indeed, once we prove this, then by the limit law for multiplication,

$$\lim_{n\to\infty} \left(a_n \cdot \frac{1}{b_n}\right) = a \cdot \frac{1}{b} = \frac{a}{b}.$$

Fix $\varepsilon > 0$. As $\lim_{n\to\infty} b_n = b$, then $\exists N \in \mathbb{N}$ st.

$$|b_n - b| < \min\left\{\frac{|b|}{2}, \frac{|b|^2 \varepsilon}{2}\right\} \quad \forall n \geq N.$$

Then, for $n \geq N$ we have

$$\left|\frac{1}{b_n} - \frac{1}{b}\right| = \frac{|b-b_n|}{|b|\cdot|b_n|} \leq \frac{2}{|b|^2}\cdot|b-b_n|$$

$$|b_n| \geq |b| - |b-b_n|$$
$$> |b| - \frac{|b|}{2} = \frac{|b|}{2}$$
$$\Rightarrow \frac{1}{|b_n|} < \frac{2}{|b|}$$

$$< \frac{2}{|b|^2}\cdot\frac{|b|^2\varepsilon}{2} = \varepsilon.$$

So $\lim_{n\to\infty} \frac{1}{b_n} = \frac{1}{b}$, as desired. $\square$

<u>Ex</u> Prove $\lim_{n\to\infty} \frac{n^3+5n+8}{3n^3+2n^2+7} = \frac{1}{3}$.

<u>Pf</u>: Note that

$$\frac{n^3+5n+8}{3n^3+2n^2+7} = \frac{1+\frac{5}{n^2}+\frac{8}{n^3}}{3+\frac{2}{n}+\frac{7}{n^3}}$$

Recall that $\lim_{n\to\infty}\frac{1}{n} = 0$ (Lecture 7). So, by the limit laws,

$$\lim_{n\to\infty}\frac{1}{n^2} = \lim_{n\to\infty}\frac{1}{n}\cdot\frac{1}{n} = 0, \qquad \lim_{n\to\infty}\frac{1}{n^3} = \lim_{n\to\infty}\frac{1}{n^2}\cdot\frac{1}{n} = 0$$

$$\Rightarrow \begin{cases} \lim_{n\to\infty}\left(1+\frac{5}{n^2}+\frac{8}{n^3}\right) = 1+0+0 \\ \lim_{n\to\infty}\underbrace{\left(3+\frac{2}{n}+\frac{7}{n^3}\right)}_{\neq 0} = \underbrace{3+0+0}_{\neq 0} \end{cases}$$

$$\Rightarrow \lim_{n\to\infty}\frac{1+\frac{3}{n^2}+\frac{8}{n^3}}{3+\frac{2}{n}+\frac{7}{n^3}} = \frac{1}{3}.$$ $\square$

Def We say that a sequence $\{a_n\}_{n\geq 1}$ is:

① <u>increasing</u> if $a_n \leq a_{n+1}$ $\forall n \geq 1$

② <u>strictly increasing</u> if $a_n < a_{n+1}$ $\forall n \geq 1$

③ <u>decreasing</u> if $a_n \geq a_{n+1}$ $\forall n \geq 1$

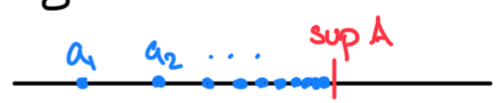
④ <u>strictly decreasing</u> if $a_n > a_{n+1}$ $\forall n \geq 1$

⑤ <u>monotone</u> if it is increasing or decreasing.

Thm If the sequence $\{a_n\}_{n\geq 1}$ is monotone and bounded, then it converges.

Pf: Case: $\{a_n\}_{n\geq 1}$ is increasing. Set

$$A = \{a_n : n \geq 1\}.$$



Then:

- $A \neq \emptyset$: since $a_1 \in A$
- A is bounded above: since $\{a_n\}_{n\geq 1}$ is bounded.

So, by the least upper bound property, $\exists\, a \in \mathbb{R}$ s.t. $a = \sup A$.

Claim: $\lim_{n\to\infty} a_n = a$. Fix $\varepsilon > 0$. Then $a - \varepsilon$ is not an upper bound for A, so $\exists\, N \in \mathbb{N}$ s.t. $a_N > a - \varepsilon$. So, for $n \geq N$,

$$\left.\begin{array}{l} a_n \geq a_N > a - \varepsilon \\ \quad \uparrow \{a_n\}_{n\geq 1} \text{ increasing} \end{array}\right\} \Rightarrow -\varepsilon < a_n - a < \varepsilon$$

$a_n \le a < a + \varepsilon$

$a = \sup A$ is an upper bound

$\Rightarrow \quad |a_n - a| < \varepsilon.$

So $\lim_{n\to\infty} a_n = a$.

Case: $\{a_n\}_{n\ge 1}$ is decreasing. Then $\{-a_n\}_{n\ge 1}$ is increasing. Apply the previous case. (Exercise.) $\square$